

Numerical Integration:-

Aim:-

$$I = \int_a^b f(x) dx = ? \quad (\text{Numerically})$$

Here $f(x)$ may be explicitly given or given as a tabulated data.

Assumed:- $f(x)$ is integrable on $[a, b]$

$$I = \int_a^b f(x) dx = \sum_{k=0}^n \lambda_k f(x_k)$$

Recall
from
12th

$$= \lambda_0 f(x_0) + \lambda_1 f(x_1) + \dots + \lambda_n f(x_n)$$

Known $\rightarrow f(x_i)$ | Unknown $\rightarrow \lambda_i$'s

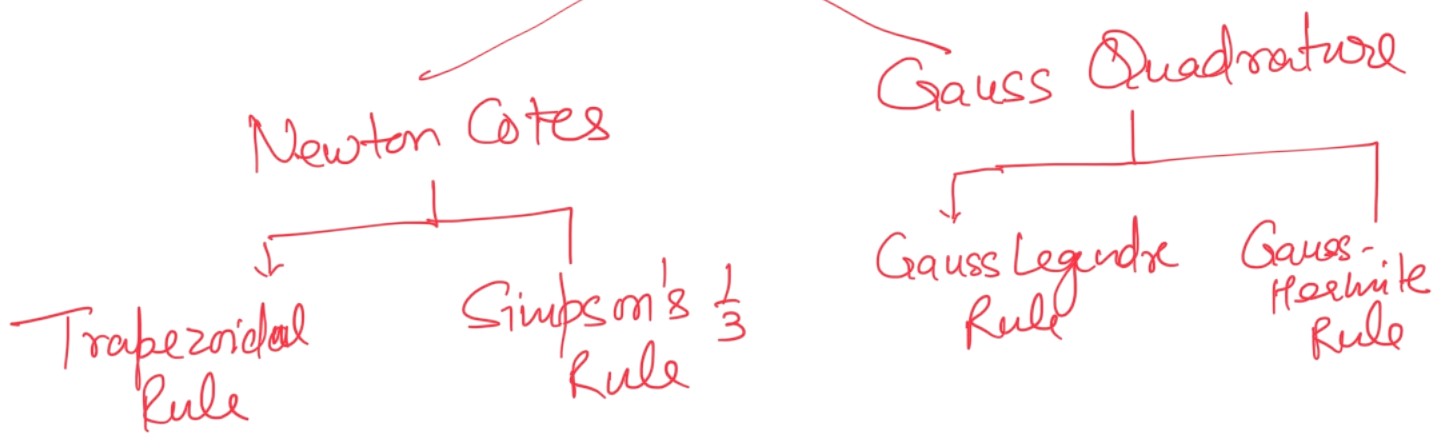
Newton-Cotes Integration Rules:-

To find. $I = \int_a^b f(x) dx$

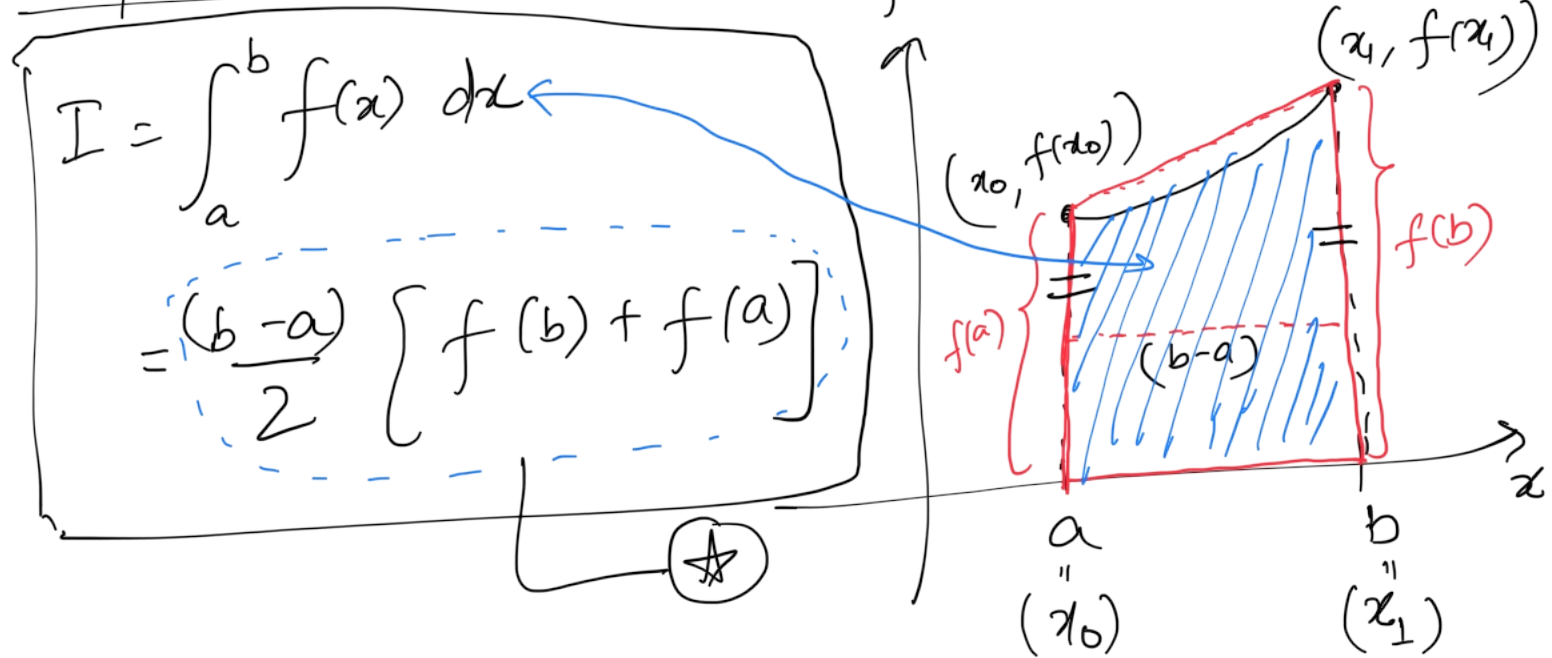
The nodes $x_0, x_1, \dots, x_n$ are equispaced.

⊛ we know that $\int_a^b f(x) dx$ defines area under the curve $y = f(x)$ above x -axis between the lines $x = a$ & $x = b$.

Integration Rules / Quadrature formula

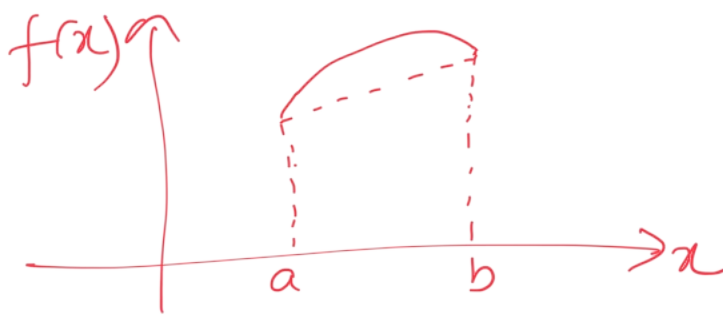


Trapezoidal Rule:- $x_0 = a$, $x_1 = b$



Remark:- Geometrically, R.H.S. of $(*)$ is the area of trapezium = $(f(a) + f(b)) \times \frac{(b-a)}{2}$

Think about it:-



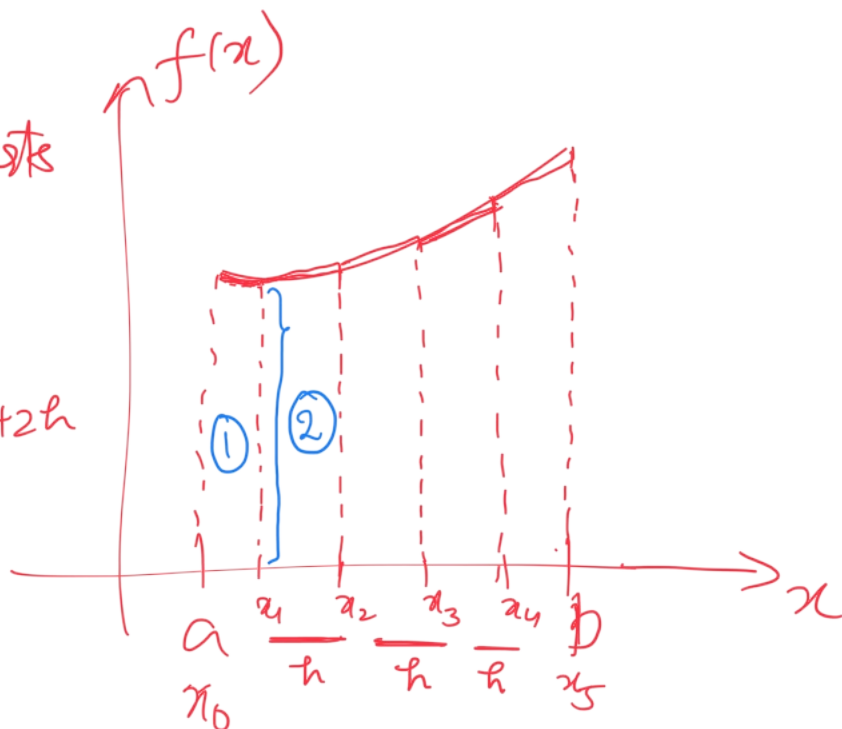
Composite trapezoidal Rule :-

divide the interval $[a, b]$ into N equal parts of length h .

$$a = x_0, x_1 = x_0 + h, x_2 = x_0 + 2h$$

$$\dots x_N = x_0 + Nh = b$$

$$h = \frac{b-a}{N}$$



$$I = \int_a^b f(x) dx = \frac{h}{2} [f(x_0) + f(x_1)] + \frac{h}{2} [f(x_1) + f(x_2)] + \dots + \frac{h}{2} [f(x_{N-1}) + f(x_N)]$$

$$\Rightarrow I = \frac{h}{2} \left[f(x_0) + 2 \{ f(x_1) + f(x_2) + \dots + f(x_{N-1}) \} + f(x_N) \right]$$

Remark:- Geometrically R.H.S. of $\textcircled{\star\star}$ is the sum of areas of N trapeziums with height h .

Ques:- Find $I = \int_0^1 \frac{1}{1+x} dx$ & $N = 8$

$$h = \frac{1-0}{8}$$

Exact solution:- $I = \ln 2 = 0.693147$

$$x: x_0 = 0, \frac{1}{8}, \frac{2}{8}, \frac{3}{8}, \frac{4}{8}, \frac{5}{8}, \frac{6}{8}, \frac{7}{8}, x_8 = 1$$

$$f(x): 1, \frac{8}{9}, \frac{8}{10}, \frac{8}{11}, \frac{8}{12}, \frac{8}{13}, \frac{8}{14}, \frac{8}{15}, 0.5$$

$$f(x) = \frac{1}{1+x} \text{ then}$$

$$I = \frac{h}{2} \left[f(x_0) + 2 \left\{ f(x_1) + f(x_2) + \dots + f(x_7) \right\} + f(x_8) \right]$$

$$I = 0.6941 \quad \checkmark$$

$$\text{Error} = |0.693147 - 0.694122| = 0.000975$$

Apply Eqⁿ (★)

$$\begin{array}{c} | \\ x_0 = 0 \end{array} \quad \begin{array}{c} | \\ x_1 = 1 \\ h = 1 \end{array}$$

$$I = \frac{h}{2} [f(a) + f(b)]$$

$$= \frac{1}{2} [1 + 0.5] = 0.75 \quad \checkmark$$

Now use $N = 2$:

$$\begin{array}{c} | \quad | \quad | \\ x_0 \quad x_1 \quad x_2 \\ 0 \quad 0.5 \quad 1 \end{array}$$

$$f(x) = 1, \frac{2}{3}, 0.5$$

$$I = \frac{h}{2} [f(x_0) + 2f(x_1) + f(x_2)] = 0.7083$$

Do it yourself:-

Ques:- Find $I = \int_1^2 \frac{dx}{5+3x}$ with $N=8$
Compare your results with exact solution.
